RESEARCH METHODOLOGY & IPR - R1UC701T

QUESTION BANK

UNIT - 1

1. Define technical writing and list the different types of documents it includes.
2. State the four characteristics (4 C's) of technical writing.
3. List the basic principles of good technical writing.
4. Name the qualities required for effective technical writing.
5. State the purpose of technical writing.
6. Identify the different types of technical reports and name their general outline.
7. Explain the different steps involved in writing a research report.
8. Describe the layout of a research report and outline the contents included in each section.
9. Discuss the statement "Report writing is more an art that hinges upon practice & experience," and narrate the various steps involved in writing such a report.
10. Explain the different types of reports and describe the general outline of a technical report.
11. Describe the difference between a technical report and a popular report, giving their respective outlines.
12. Discuss the precautions a researcher should keep in mind while preparing a good research report

UNIT - 2

1. Analyze the various sources of a research problem and justify how prior research and intellectual curiosity contribute to identifying a good research problem.

2. Examine the criteria for selecting a good research problem and relate how factors such as feasibility, originality, and researcher's competence influence the outcome of a study.

3. Differentiate between quantitative and qualitative errors in selecting a research problem, and analyze their impact on the validity of research findings.

4. Compare the scope and objectives of a research problem, and examine how they collectively guide the direction of a research study.

5. Classify the different types of research problems (theoretical, applied, and action research) and analyze the situations in which each type would be most appropriate.

6. Examine the common errors while selecting a research problem, and suggest measures to minimize such errors.

7. Evaluate the six ways to avoid plagiarism and propose measures an institution could adopt to prevent it.

8. Design a set of research ethics guidelines and justify the importance of each principle included.

9. Critique the four types of plagiarism based on their severity and formulate a strategy to detect them.

10. Construct a scenario illustrating accidental plagiarism and propose suitable corrective measures.

11. Evaluate how research ethics influences the credibility and trustworthiness of research findings.

12. Formulate a comprehensive plagiarism and research ethics policy suitable for a university or research institute.

UNIT 3

1. Evaluate the different literature study approaches and justify which is most suitable for a systematic review.

2. Critique the process of conducting an effective literature review and propose improvements to it.

3. Assess the typical structure of a literary analysis and justify the importance of each stage.

4. Evaluate the role of critical appraisal and citation tracking in an effective literature review.

5. Judge the strengths and limitations of narrative review versus systematic review approaches.

6. Critically examine how literature analysis contributes to formulating a strong research problem.

7. Evaluate the quantitative, qualitative, and pragmatic approaches to investigating a research problem.

8. Assess the suitability of different primary data collection methods for a given research scenario.

9. Critique the techniques and precautions used in interpreting research data.

10. Evaluate how the choice of research instrument affects the reliability of data collected.

11. Judge the effectiveness of descriptive versus inferential analysis in drawing research conclusions.

12. Critically assess the guidelines for developing a valid and reliable research instrument.